

Lifecycle Justification

1. Introduction

The University Course Registration System is designed to help students register for courses, view their schedules, and manage their academic plans efficiently. It also allows administrators to manage courses, instructors, and student records.

Selecting an appropriate software development lifecycle is essential to ensure that the system is flexible, efficient, and capable of meeting user requirements.

2. Justification for Using Agile Methodology

The Agile methodology is considered the most suitable approach for developing this system due to the following reasons:

2.1 Changing Requirements

In a university environment, requirements frequently change. For example:

- New courses may be introduced each semester
- Registration rules and policies may be updated
- Academic structures may evolve over time

Agile supports such changes by allowing continuous updates without restarting the development process.

2.2 Continuous User Feedback

The system involves multiple users, mainly students and administrators. Agile enables:

- Early access to system features

- Continuous feedback after each sprint

This ensures that the system aligns with actual user needs.

2.3 Incremental Delivery

Agile delivers the system in small, manageable increments. For example:

- Sprint 1: User Registration and Login
- Sprint 2: Course Browsing and Selection
- Sprint 3: Schedule Management

Each increment provides a working part of the system, improving usability and reliability.

2.4 Risk Reduction

Agile helps in identifying and resolving issues early in the development process. This includes:

- Early detection of bugs
- Immediate correction of design issues

As a result, the overall project risk is significantly reduced.

2.5 Improved Team Collaboration

Agile promotes effective communication among team members through:

- Frequent meetings
- Clear task distribution

This enhances teamwork and productivity.

2.6 Faster Delivery

Agile ensures that a functional version of the system is available early, with continuous improvements over time. This reduces waiting time and increases user satisfaction.

3. Comparison with Traditional Models

Compared to the Waterfall model, Agile provides greater flexibility and adaptability.

Aspect	Waterfall	Agile
Flexibility	Low	High
User Feedback	Late	Continuous
Handling Changes	Difficult	Easy
Delivery	End of project	Incremental

4. Conclusion

Agile methodology is the most appropriate choice for the University Course Registration System as it supports flexibility, continuous improvement, and active user involvement. It ensures that the system evolves according to real user needs while maintaining high quality and efficiency.